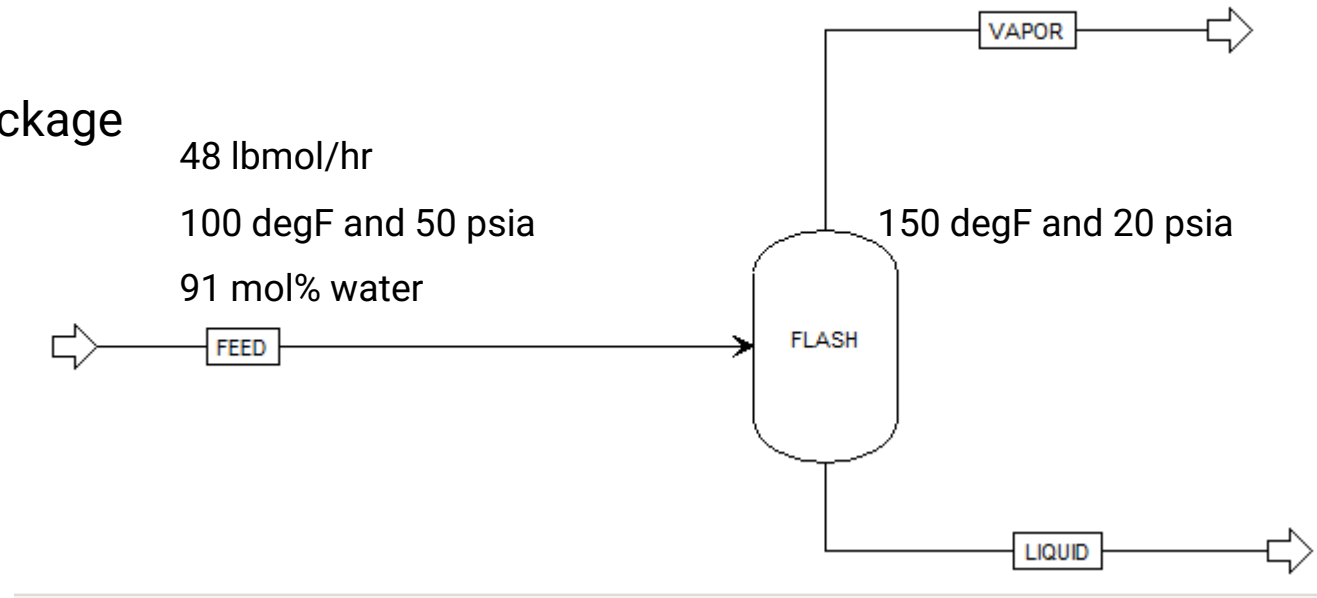


Ejercicio 3: Conectar un Modelo de ASPEN PLUS en FOQUS

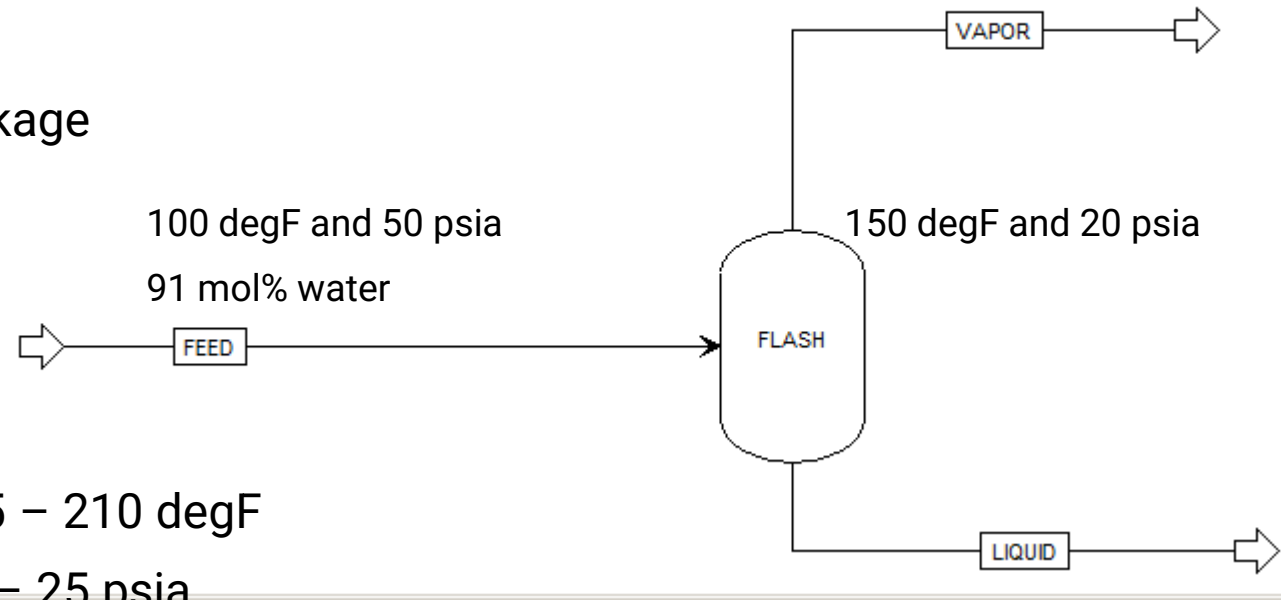
Ejercicio 3: Flash Drum Aspen Model in FOQUS

- We have an Aspen flowsheet setup for a flash operation. The feed consists of 91 mol% water and 9 mol% ethanol at 100 degF and 50 psia. 48 lbmol/hr of the mixture is fed to the flash tank which is at 150 degF and 20 psia. We want to know the composition of the VAPOR and LIQUID streams.
- Add Water and Ethanol
- Select UNIFAC Property Package



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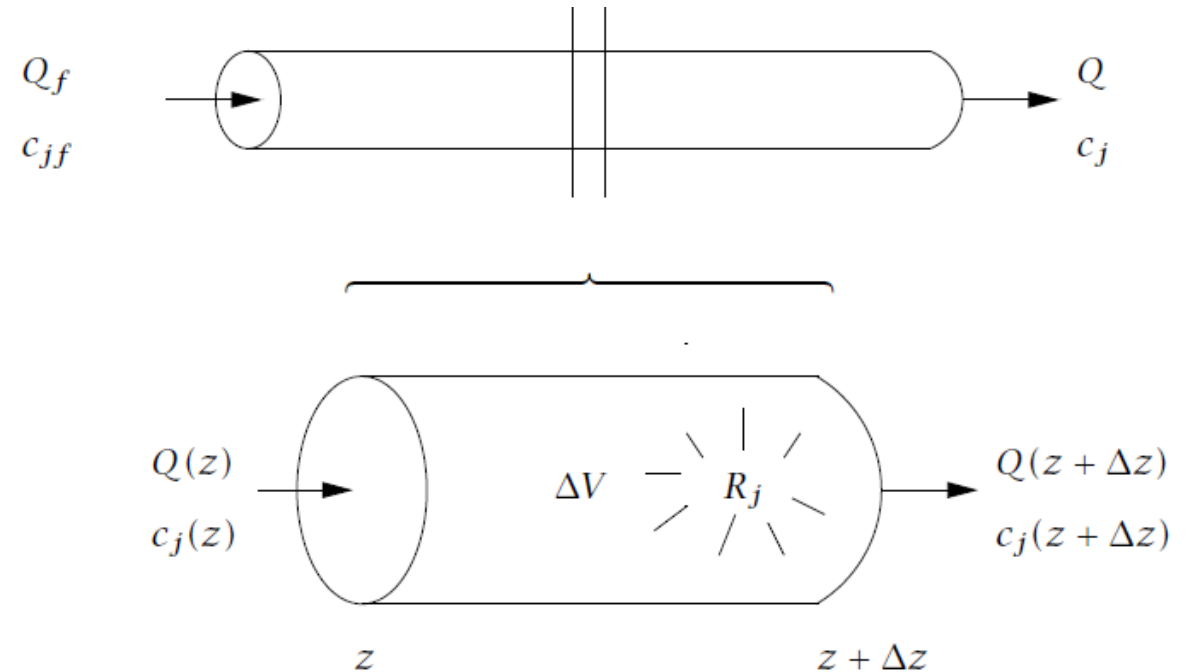


- Analyze Txy plot
- Lower/upper bound TF = 185 – 210 degF
- Lower/upper bound Tp = 18 – 25 psia

Ejercicio 4: Desarrollar Modelos Sustitutos (surrogate models) de un Plug Flow Reactor

Main Assumptions

- Ideal flow assumption,
- Fluid is well mixed in the r
- A thin disk for the reactor
- Constant cross section, A_c
- Steady state



Ejercicio 4. PFR Reactor Model

- Material balance for the volume element

$$\frac{\partial (c_j \Delta V)}{\partial t} = c_j Q \Big|_z - c_j Q \Big|_{z+\Delta z} + R_j \Delta V$$

- Dividing by ΔV and taking the limit as ΔV goes to zero

$$\underbrace{\frac{\partial c_j}{\partial t}}_{\text{accumulation}} = - \underbrace{\frac{\partial (c_j Q)}{\partial V}}_{\text{convection}} + \underbrace{R_j}_{\text{reaction}}$$

- Constant cross section A_c , then the velocity is related to the volumetric flow rate, and the axial length is related to the tube volume.

$$\frac{\partial c_j}{\partial t} = - \frac{\partial (c_j v)}{\partial z} + R_j$$

Ejercicio 4. PFR Reactor Model

$$\frac{d(c_j Q)}{dV} = R_j$$

- Time derivative to zero

- The product is the total molar flow of component j.

$$N_j = c_j Q$$

$$\frac{dN_j}{dV} = R_j$$

Ideal Gas Equation of State

- Molar concentrations, c_j

- Volumetric flow rate, Q

$$\sum_j c_j = \frac{P}{RT}$$

$$\frac{\sum_j N_j}{Q} = \frac{P}{RT}$$

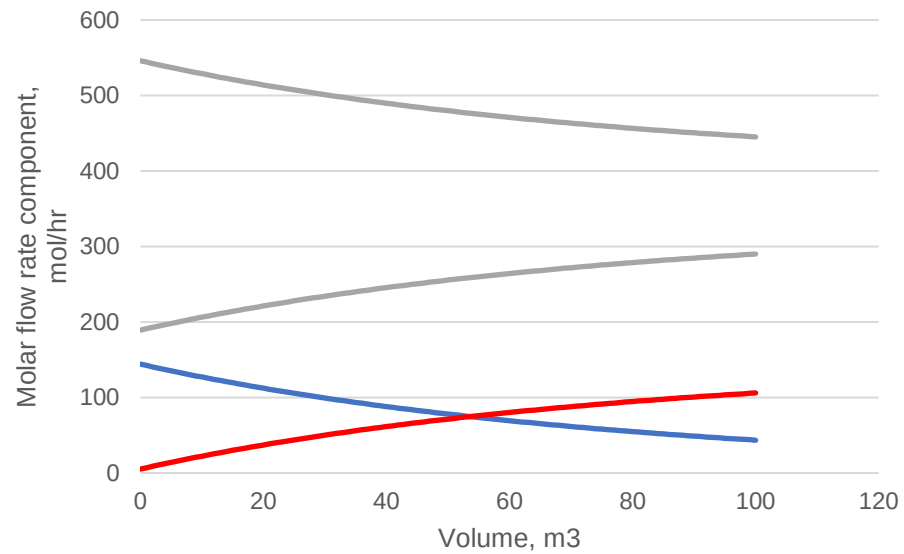
$$Q = \frac{RT}{P} \sum_j N_j$$

$$c_j = \frac{N_j}{Q} = \frac{P}{RT} \frac{N_j}{\sum_j N_j}$$

- Molar flows, N_j

- Evaluate concentrations in the reaction rate expression



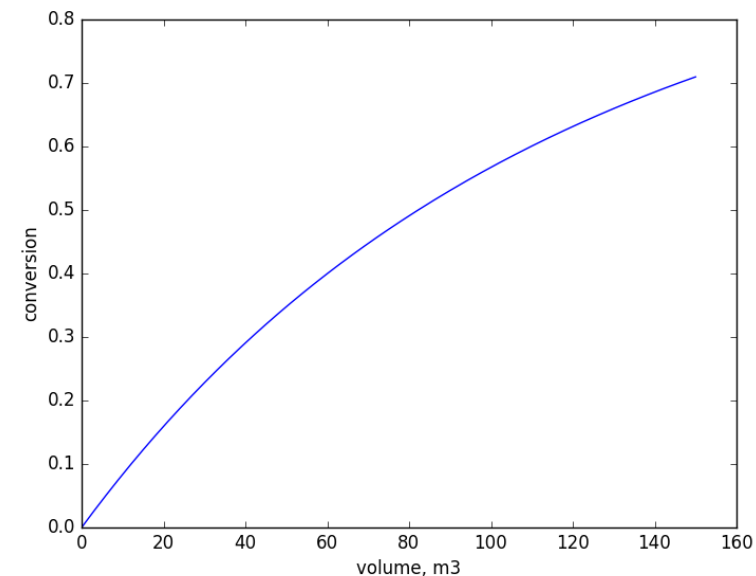
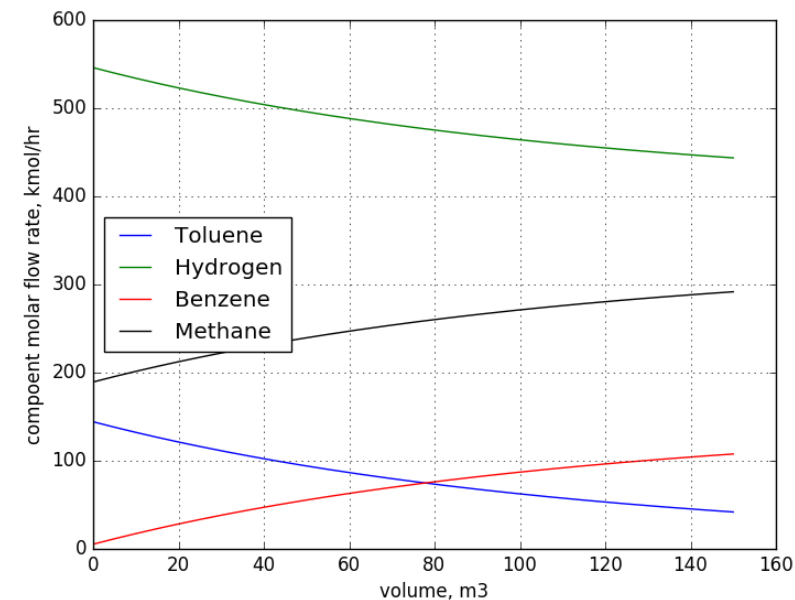
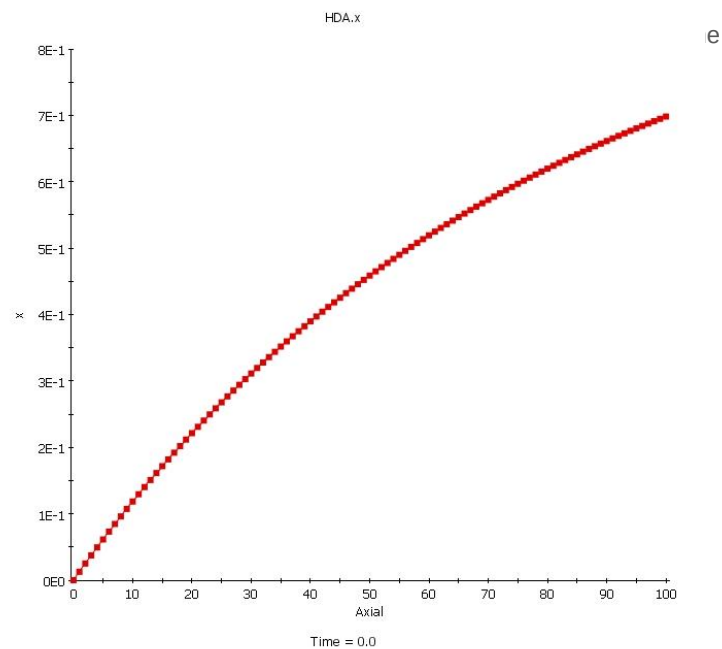


Results

← gPROMs

VS

Python →



Ejercicio 4 PFR Reactor

- Crear un nodo en FOQUS (Nombre PFR_REACTOR).
- Añadir variables de entrada y salida (hint: variable size=0)

Inputs

	Name	Value	Unit	Type	Default	Min	Max
1	benzene	3.4774		float	0.0	0.0	10.0
2	conversion	0.8762		float	0.0	0.0	1.0
3	h2	544.94		float	0.0	300.0	700.0
4	methane	135.11		float	0.0	0.0	400.0
5	pressure	24.477		float	0.0	10.0	40.0
6	temperature	678.67		float	0.0	400.0	750.0
7	toluene	228.51		float	0.0	0.0	500.0

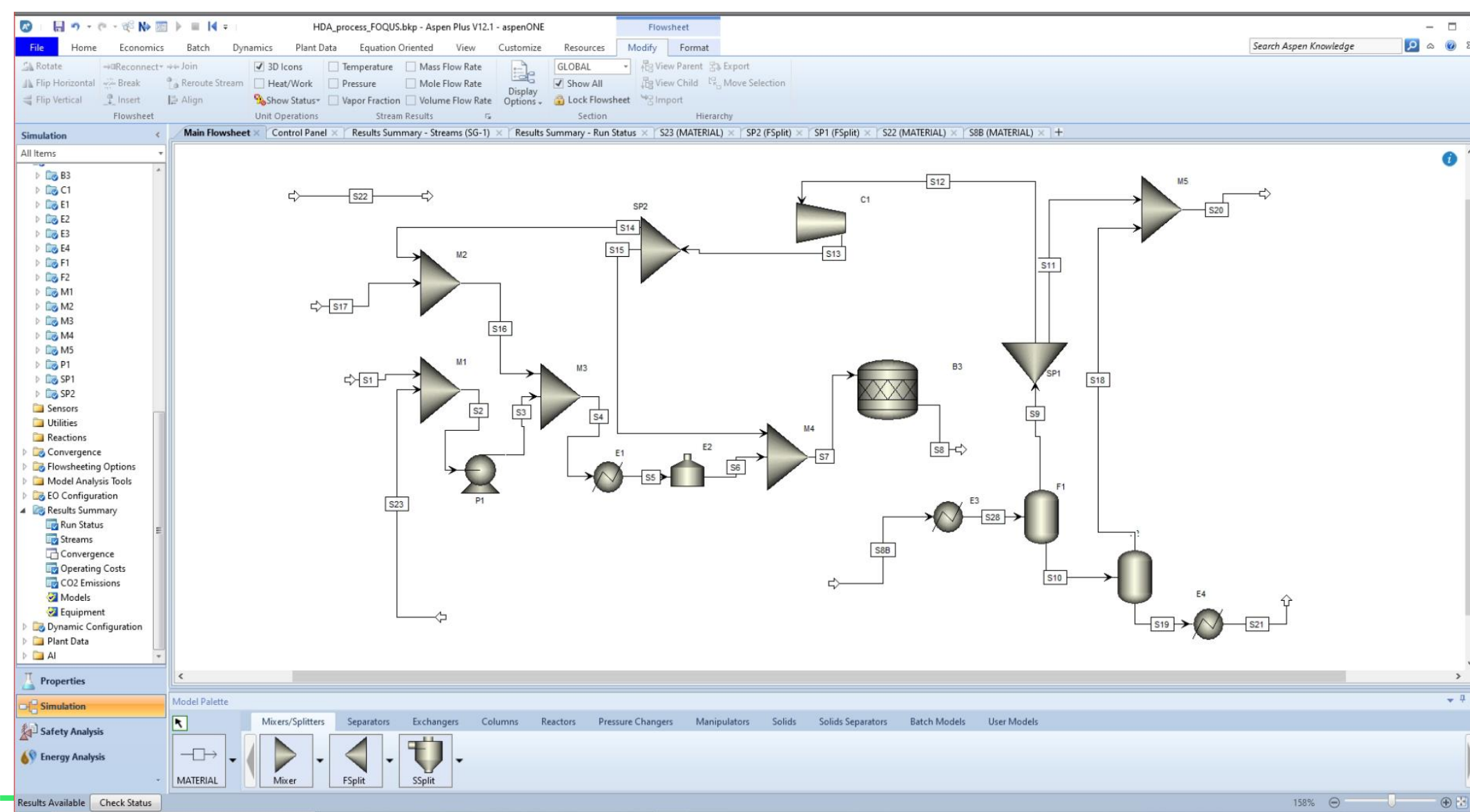
Outputs

	Name
1	Ntotal
2	outlet_C6H6
3	outlet_CH4
4	outlet_H2
5	outlet_Toluene
6	Qflow
7	volume

Ejercicio 4 PFR Reactor

- Agregar Python script del reactor PFR en el node script.
- Aplicar cambios y ejecutar la simulación.
- Analizar:
 - Análisis de sensibilidad de un parámetro(Temperatura, presión y conversion)
 - Análisis de sensibilidad multiparamétrico.
 - Desarrollar un modelo subrogado.

Ejercicio 5 HAD Process Optimization in FOQUS



AllianzEnergy
Consultores en Energías Renovables

Costing Block

FOQUS - C:\Users\usuario\Desktop\Curso_optimizacion\Module_4\Ejercicio_5_HDA\Ejercicio3_PFR_REACTOR_opt.foqus - Last saved: 2022-03-05T00:21:02

Session | Flowsheet | Uncertainty | Optimization | OUV | SDOE | Surrogates | Settings | Help

Flowchart showing a process flow: HDA (HDA_foqus) and PFR_REACTOR are connected to a Costing block. The flow is bidirectional between HDA and PFR_REACTOR, and unidirectional from both to Costing.

Node Edit Panel:

- Apply | Revert | Run (this node only for testing) | Stop Run
- Variables | Position | Node Script
- Name: PFR_REACTOR | Visible
- Error Status
- Code: 0
- Message: Finished Normally
- Model
- Type: None | Model:
- Input Variables
- Output Variables

Simulation Results Table:

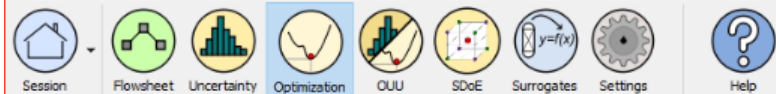
Name	Value	Unit	Type	Description	Tags
1 Ntotal	827.4202901100001		float		
2 outlet_C6H6	223		int		
3 outlet_CH4	130.5671491041371		float		
4 outlet_H2	455.14543410586265		float		
5 outlet_Toluene	18.099630105863064		float		
6 Qflow	179439.95336437583		float		
7 volume	65.0		float		

Finished in 00:01:54

The simulation completed successfully.

Running Single Simulation... Elapsed Time: 00:01:53

Windows Taskbar: 19°C Lluvia, 12:28 AM, 3/5/2022



Variables Samples Objective/Constraints Solver Run

Start Stop Update delay (ms): 500

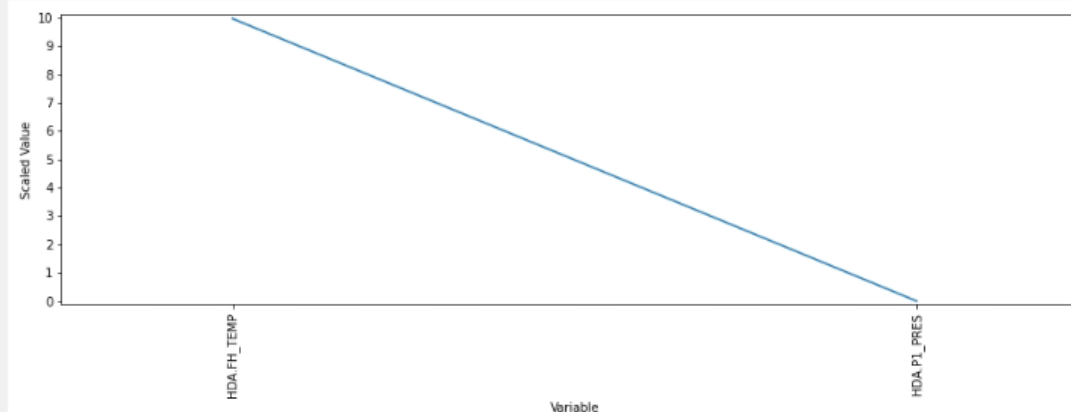
Optimization Solver Messages

29, Total Elapsed Time 6447s, Obj: 920551.2131590546
 30, Total Elapsed Time 6566s, Obj: 920551.2131590546
 31, Total Elapsed Time 6687s, Obj: 920551.2131590546
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 36, Total Elapsed Time 7353s, Obj: 920539.0905866745
 37, Total Elapsed Time 7473s, Obj: 920539.0905866745
 38, Total Elapsed Time 7789s, Obj: 920539.0905866745
 39, Total Elapsed Time 7932s, Obj: 920539.0905866745
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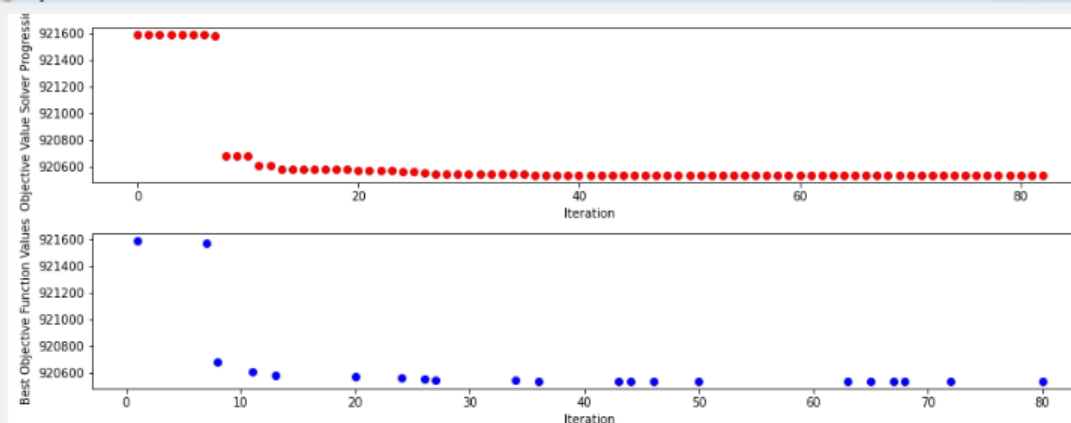
ITERATION 83: 2/6 Err: 0 TOTAL Complete: 500 Err:60

Clear

Best Solution Parallel Coordinate Plot



Objective Function Plot



Optimization Running. Elapsed Time: 07:07:16